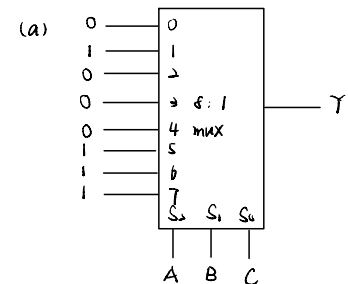
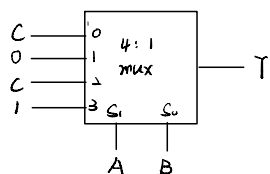


1. 真值表如下:

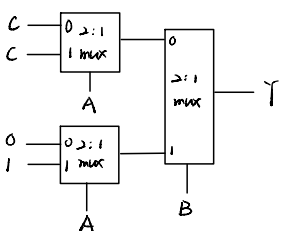
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



(b) 4:1 mux:

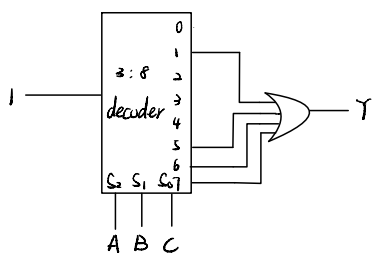


2:1 mux:



$$(c) Y = \sum m(1, 5, 6, 7) = \prod M(0, 2, 3, 4)$$

电路如下:



2. 卡诺图如下:

CD \ AB	00	01	11	10
00	1	1	0	1
01	0	0	0	1
11	0	0	0	0
10	1	0	0	1

$$\text{故 } F(A, B, C, D) = A'B'C' + AB'C' + B'CD'$$

$$= (A' + B' + C)(A + C + D')(C' + D')$$

$$(B' + C' + D)$$

3. 卡诺图如下:

Y \ AB	00	01	11	10
0	1	1	0	0
1	1	1	0	0

$$Y = A'$$

4. 卡诺图如下:

F \ AB	00	01	11	10
00	0	1	0	1
01	1	1	1	1
11	0	0	1	X
10	X	1	X	X

$$F(A, B, C, D) = A'BD' + AB' + C'D + ABC$$